

Von Neumann Entropy of Gaussian States



1. Von-Neumann (VN) Entropy: $S(\rho) = -\text{Tr}(\rho \log \rho) = -\sum_i \lambda_i \log \lambda_i$

2. For Gaussian states, VN entropy depends on covariance matrix

3. Symplectic Eigenvalues: For an N mode covariance matrix $\mathbf{V}$, there exists a symplectic matrix $\mathbf{S}$ s.t. $\mathbf{V} = \mathbf{S}\mathbf{V}^\oplus\mathbf{S}^T$, $\mathbf{V}^\oplus := \bigoplus_{k=1}^N \nu_k \mathbf{I}_2$

4. Symplectic matrix $\mathbf{S}$: $\mathbf{S}\mathbf{\Omega}\mathbf{S}^T = \mathbf{\Omega}$ where $\mathbf{\Omega} := \bigoplus_{k=1}^N \omega = \begin{pmatrix} \omega & & \\ & \ddots & \\ & & \omega \end{pmatrix}$, $\omega := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

5. Symplectic spectrum equals standard eigenspectrum of $|i\mathbf{\Omega}\mathbf{V}|$

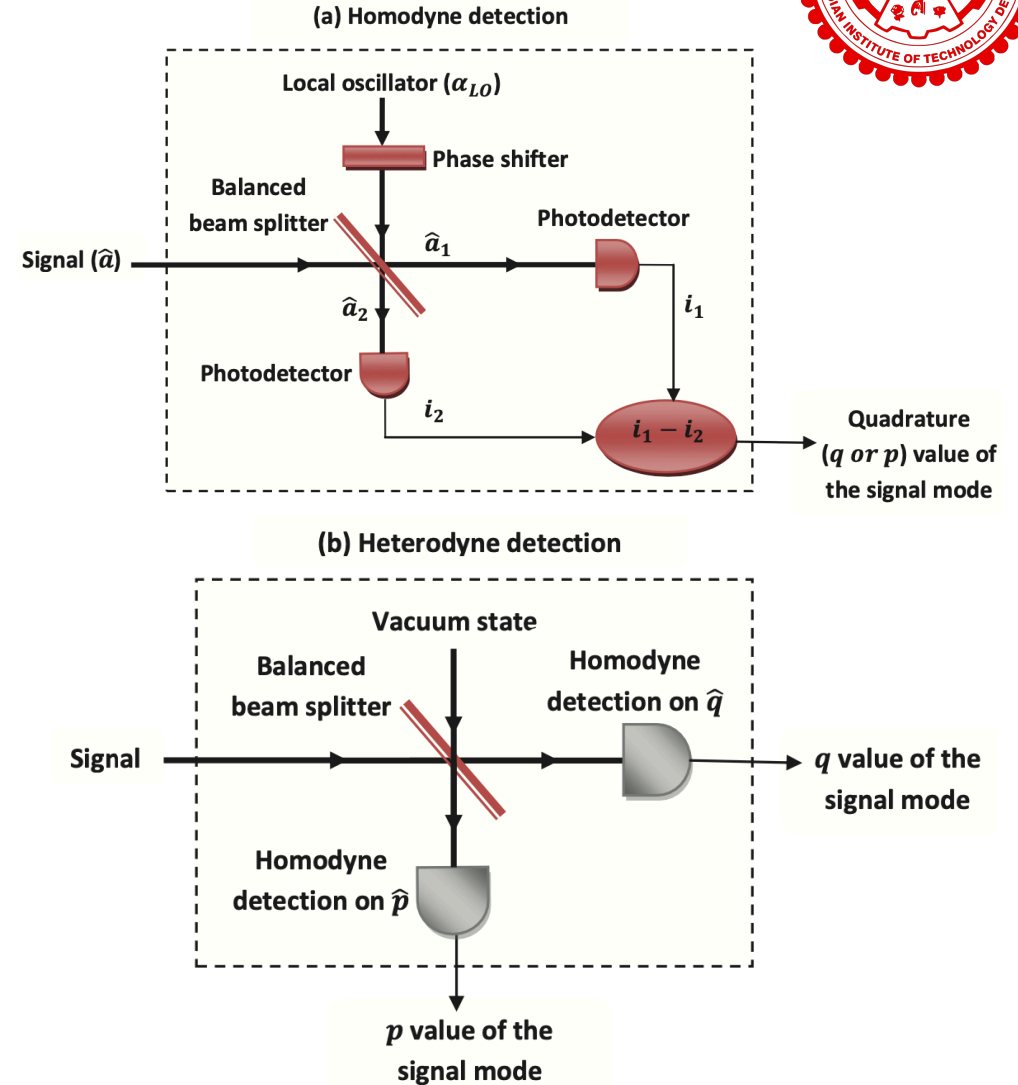
6. VN Entropy:

$$S(\hat{\rho}) = \sum_{k=1}^N h(\nu_k), \text{ where } h(x) := \left(\frac{x+1}{2}\right) \log \left(\frac{x+1}{2}\right) - \left(\frac{x-1}{2}\right) \log \left(\frac{x-1}{2}\right).$$

Quadrature Measurement



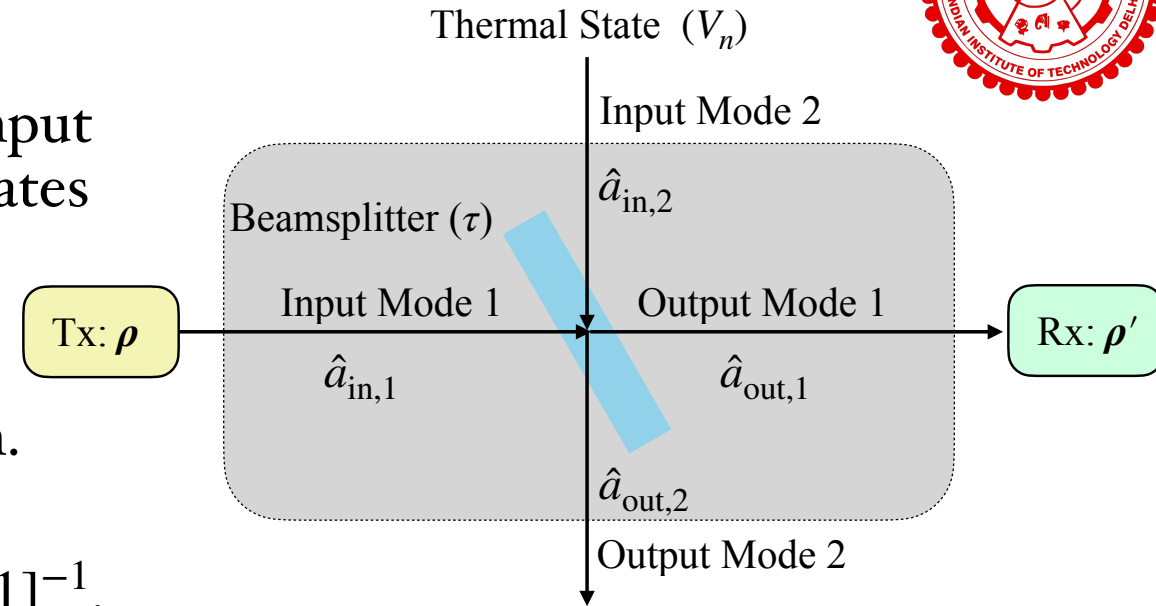
1. **Homodyne Detection:** LO field: $|\alpha_{LO}\rangle$.
2. O/P mode of balanced BS: $\hat{a}_1 = (\alpha_{LO} + \hat{a})/\sqrt{2}$ and $\hat{a}_2 = (\alpha_{LO} - \hat{a})/\sqrt{2}$
3. Photodetectors output currents:
 $\hat{i}_1 \propto \hat{n}_1 = \hat{a}_1^\dagger \hat{a}_1 = \left(\alpha_{LO}^* + \hat{a}^\dagger \right) (\alpha_{LO} + \hat{a})/2$ and
 $\hat{i}_2 \propto \hat{n}_2 = \hat{a}_2^\dagger \hat{a}_2 = \left(\alpha_{LO}^* - \hat{a}^\dagger \right) (\alpha_{LO} - \hat{a})/2$
4. Difference of photocurrent:
 $\hat{i}_1 - \hat{i}_2 \propto \left(\alpha_{LO}^* \hat{a} + \alpha_{LO} \hat{a}^\dagger \right)$.
5. Considering $\alpha = |\alpha_{LO}| \exp(i\Theta)$, then $\hat{q}(\hat{p})$ can be measured by setting the local oscillator's phase as $\Theta = 0$ ($\Theta = \pi/2$).
6. **Heterodyne Detection:** Combine with Vacuum mode on balanced BS + homodyne detection on conjugate quadratures (with additional noise).



Quantum Channel

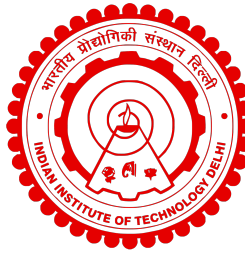


1. **Gaussian lossy channel:** Transform input Gaussian states into output Gaussian states
2. Channel Parameters:
 1. **Channel transmissivity :**
 $0 \leq \tau \leq 1$ models the attenuation.
 2. **Thermal environment noise:**
 $V_n = 2\bar{n} + 1$, $\bar{n} = [\exp(hf/k_B T_e) - 1]^{-1}$.
3. **Beamsplitter model:** Input state gets mixed with the thermal environment noise
4. Maintains the commutator relations as required by the laws of QM

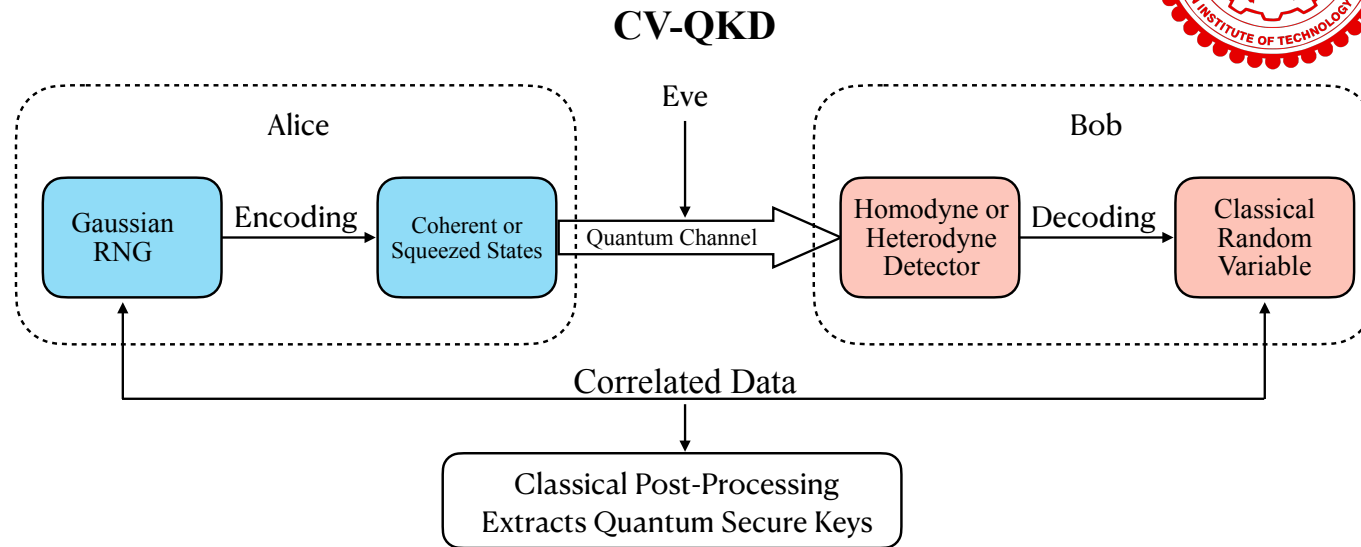


$$\begin{bmatrix} \hat{a}_{out,1} \\ \hat{a}_{out,2} \end{bmatrix} = \begin{bmatrix} \sqrt{\tau} & \sqrt{1-\tau} \\ -\sqrt{1-\tau} & \sqrt{\tau} \end{bmatrix} \begin{bmatrix} \hat{a}_{in,1} \\ \hat{a}_{in,2} \end{bmatrix}$$

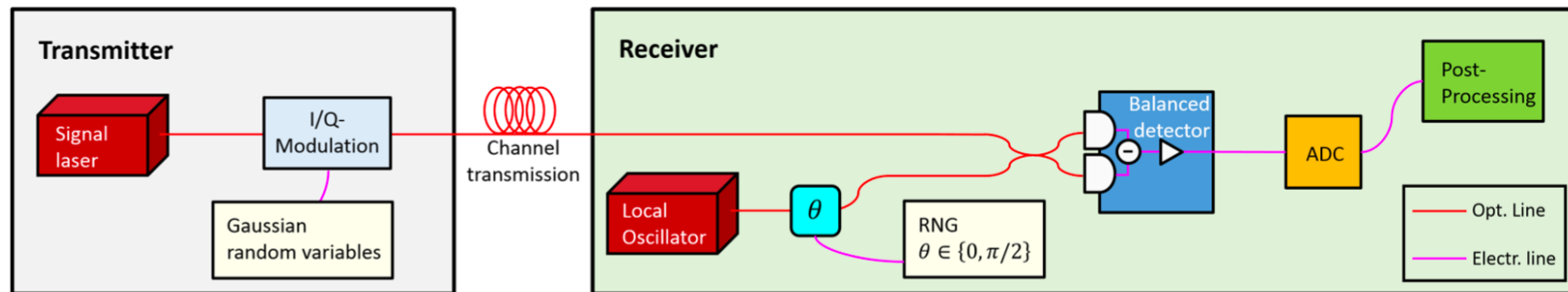
Continuous Variable QKD



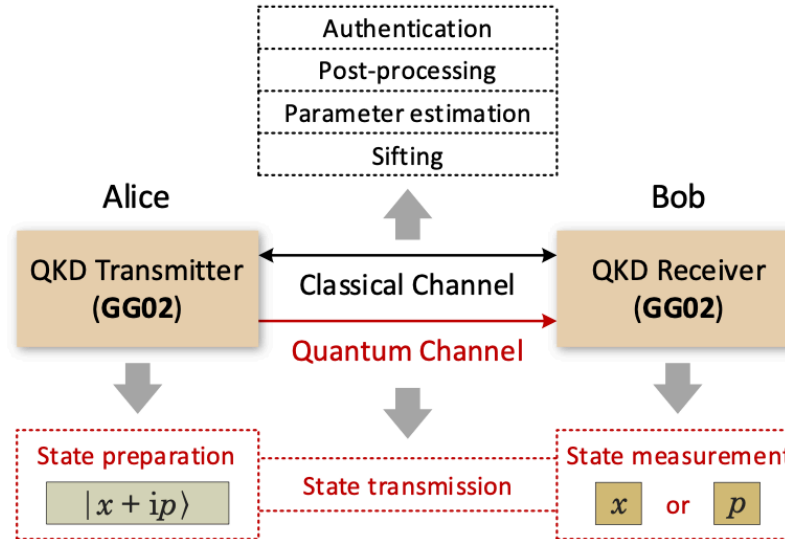
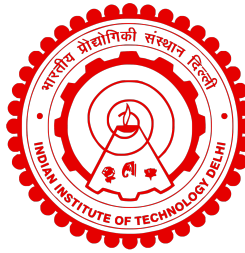
1. Encode key information in quadrature of coherent states
2. Uses lasers and homodyne/heterodyne detectors
3. Easy integration with current telecom technology



Coherent-State CV-QKD / homodyne detection

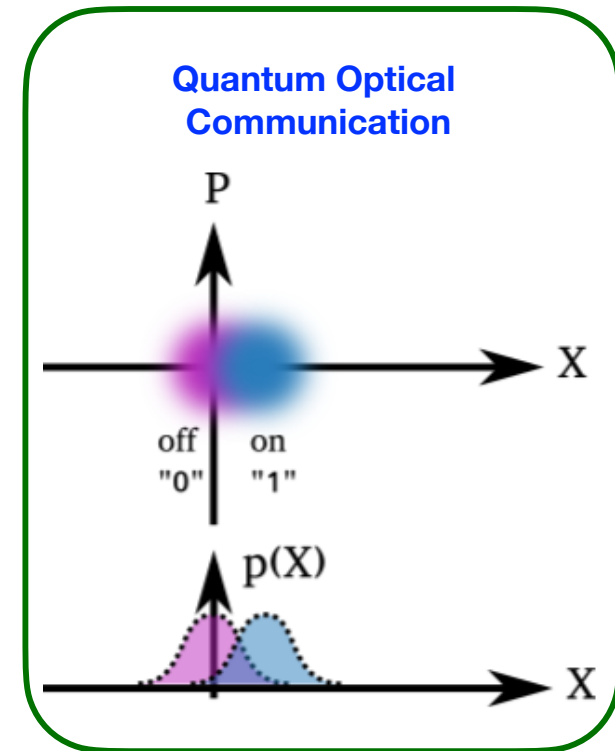
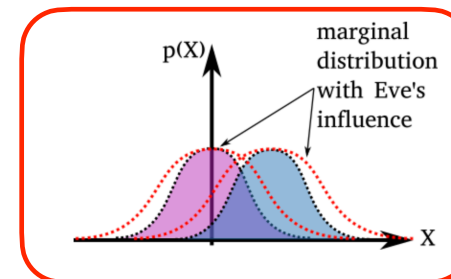
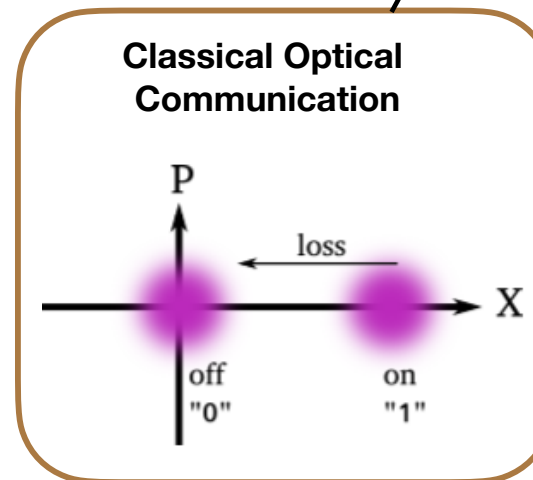
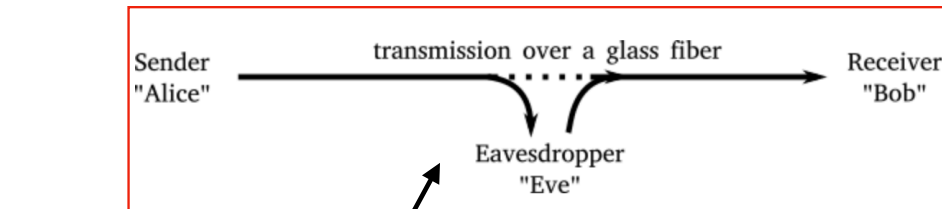
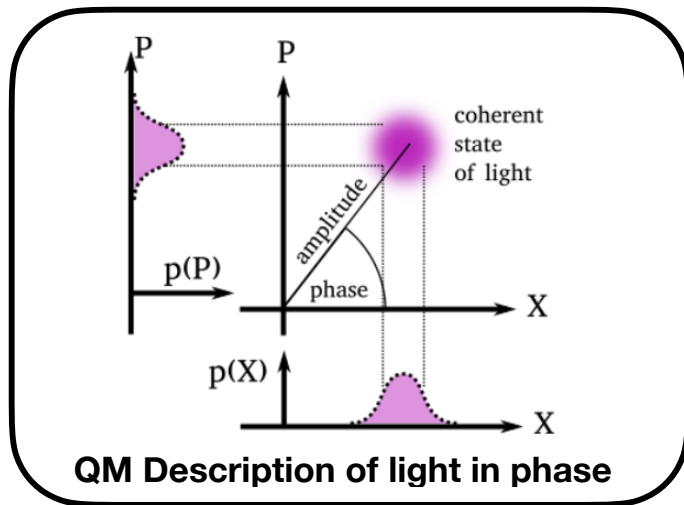
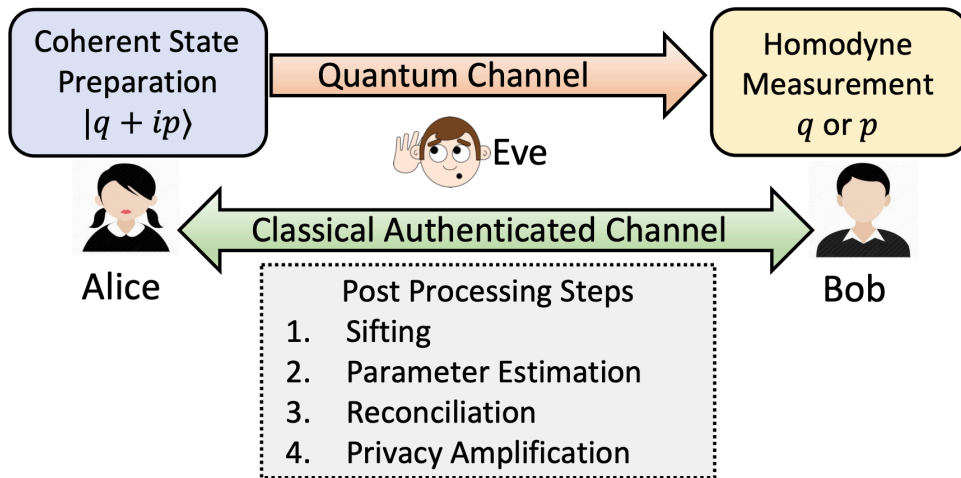
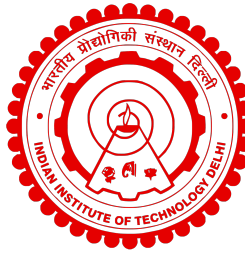


CV-QKD Protocol



Alice											
1	Raw Gaussian key (a_q)	0.9	2.4	1.3	2.1	0.5	4.1	1.1	0.3	3.6	0.2
2	Raw Gaussian key (a_p)	1.2	0.9	5.1	2.7	1.5	3.1	0.6	4.3	2.8	0.1
3	Coherent state preparation (mean)	(0.9, 1.2)	(2.4, 0.9)	(1.3, 5.1)	(2.1, 2.7)	(0.5, 1.5)	(4.1, 3.1)	(1.1, 0.6)	(0.3, 4.3)	(3.6, 2.8)	(0.2, 0.1)
Bob											
4	Detection quadrature	$\hat{p}$	$\hat{p}$	$\hat{q}$	$\hat{p}$	$\hat{q}$	$\hat{q}$	$\hat{p}$	$\hat{q}$	$\hat{q}$	$\hat{p}$
5	Detected Gaussian key	1.2	0.9	1.3	2.7	0.5	4.1	0.6	0.3	3.6	0.1
Classical post-processing											
6	Sifted Gaussian key	1.2	0.9	1.3	2.7	0.5	4.1	0.6	0.3	3.6	0.1

CV-QKD vs Classical Communication



CV-QKD Security Analysis



1. Two equivalent CVQKD schemes:
 1. **Prepare and measure (PM)** : Alice uses coherent states for encoding and transmits to Bob
 2. **Entanglement based (EB)**: Alice generates EPR states and performs homodyne detection
2. Reconciliation Schemes:
 1. **Direct Reconciliation (DR)** : Alice's raw key data is used as reference for error correction
 2. **Reverse Reconciliation (RR)** : Bob's raw key data is used as reference for error correction
3. Attack Models:
 1. **Individual Attack**: Eve manipulates and measure the transmitted states independently and identically.
 2. **Collective Attack**: Eve manipulates the transmitted states independently and identically, but measure them jointly.
 3. **Coherent Attack**: Eve manipulates and measures the transmitted states jointly. Usually coherent attack is strongest one.



Secret Key Rate

1. **Input-Output relation:** $\hat{a}_B = \sqrt{T}\hat{a}_A + \sqrt{1-T}\hat{a}_E$
2. **Quadrature Relation for Homodyne Measurement:** $x_B = \sqrt{T}x_A + \sqrt{1-T}x_0 = \sqrt{T}(x_A + \sqrt{\chi_{line}}x_0)$
where $\chi_{line} = (1-T)/T = 1/T - 1$ is total channel added noise.
3. For Lossy Transmission with excess noise ϵ (laser phase noise, etc.): $x_B = \sqrt{T}(x_A + \sqrt{\chi_{line}}x_0)$ where $\chi_{line} = (1-T)/T + \epsilon = 1/T - 1 + \epsilon$ in shot noise units (SNU)
4. Variance of Bob's r.v. $\text{Var}(x_B) = V_B = T(V + 1 + \chi_{line})$, Here V is the modulation variance of Alice
5. **SKR for RR:** $R = \beta I_{AB} - S_{BE}$ where β is reconciliation efficiency and
6. I_{AB} : **Classical Shannon's mutual information** between the random variables at Alice and Bob:
$$I_{AB} = \frac{1}{2} \log_2 \left(1 + \frac{TV}{T(1 + \chi_{line})} \right)$$
7. S_{BE} : **Holevo information** between Eve's state and Bob's measurements (Bounds the maximum information leaked to Eve allowed by quantum mechanics)

SKR Analysis -II



1. The quantum mutual information between between Bob and Eve, S_{EB} is bounded by Holevo quantity: $S_{BE} = \chi_{BE} = S(E) - S(E|m_B)$ where $S(E)$ is the von Neumann entropy of the eavesdropper's state ρ_E , and $S_{E|m_B}$ is von Neumann entropy conditional on Bob's measurement result and determined by the detection methods.
2. Based on the fact that the eavesdropper Eve is able to purify the binary quantum system ρ_{AB} :
 $\chi_{BE} = S(\rho_{AB}) - S(\rho_{AB}|m_B)$
3. Calculations of $S(\rho_{AB})$ and $S(\rho_{AB|m_B})$ can be simplified using the symplectic eigenvalues of the covariance matrix Σ_{AB} and $\Sigma_{AB|m_B}$ respectively, corresponding to the state ρ_{AB} and $\rho_{AB|m_B}$.
4. Covariance matrix Σ_{AB} can be estimated from the modulation and detection data:
5. $\Sigma_{AB} = \begin{bmatrix} \gamma_A I & \sigma_{AB} \\ \sigma_{AB} & \gamma_B \end{bmatrix} = \begin{bmatrix} V I_2 & \sqrt{T(V^2 - 1)} \sigma_z \\ \sqrt{T(V^2 - 1)} \sigma_z & V_B I_2 \end{bmatrix}$ where $V_B = T(V + 1 + \chi)$
6. $\Sigma_{AB|m_B} = \gamma_A - V_B^{-1} \sigma_{AB} \Pi \sigma_{AB}^T$ where $\Pi = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$



SKR Analysis -III

1. The symplectic eigenvalues of a covariance matrix Σ can be achieved by calculating the eigenvalues of $i\Omega\Sigma$.

2. The Holevo Information in RR is given by: $\chi_{BE} = \sum_{i=1}^2 h(\lambda_i) - h(\lambda_3)$ where $\lambda_{1,2}$ are the symplectic eigenvalues of CM Σ_{AB} and λ_3 is the symplectic eigenvalue of CM $\Sigma_{AB|m_B}$ and

$$h(x) = \frac{x+1}{2} \log_2 \frac{x+1}{2} - \frac{x-1}{2} \log_2 \frac{x-1}{2}$$

3. The SKR is given by: $R = \beta I_{AB} - \chi_{BE}$ where β is the reconciliation efficiency

Thank You